

Figure 3 Gate

Two Framings of the Single-Deletion Multimodal Figure, as Mockups

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Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 The decision ✗

Figure 3 covers single-gene deletions read out in three modalities: the transcriptome (Kemmeren 2014, 1,484 deletions), the proteome (Messner 2023, 4,476) and morphology (Ohya 2005, CalMorph, 4,718). Two framings are open and they send the next compute to different places. Framing A is a conditioning figure: one modality is measured and another is predicted from it, which is an imputation claim and needs the conditioning measurement for every strain the model is applied to (fig. 1). Framing B is a genotype-only figure: what the deletion alone predicts against each replicate ceiling and against linear and nearest-neighbor baselines (fig. 2). The choice is A, B, or neither; figs. 3 and 4 hold the supporting panels, none of which repeats a main panel, and table 1 lists what each panel lacks. The joint-training round v19 is a null on both registered tests. The conditioned round v20 is in flight at 2 of 12 split seeds and about 160 of 1,200 epochs, so framing A currently rests on the ridge triangle and on a partial run. On file, the best nearest-neighbor baseline is within 0.01 of the best model score on expression and above it on the proteome and on morphology, each a maximum over keys selected on validation (fig. 1b). SRC: `experiments/019-simb-multimodal/results/joint_checkpoint_readout.json`

Status marks. Every panel letter carries the status glyph used for section headings here: ✓ ready, the panel can be drawn now from committed results; ■ partial, or waiting on an author decision; ✗ not done, a placeholder. The marks are mockup annotations on their own draw.io layer and are switched off with `FIG3_STATUS_MARKS=0`. A *split seed* is a data partition: each seed is a different train, validation and test split, run with one initialization seed.

2 The mockups ✗

Result files are under `experiments/019-simb-multimodal/results/` unless a path is given. Each caption gives, per panel, what is plotted, the statistic and its scoring rule, the sample size and the status; the table under each figure gives the files and keys.

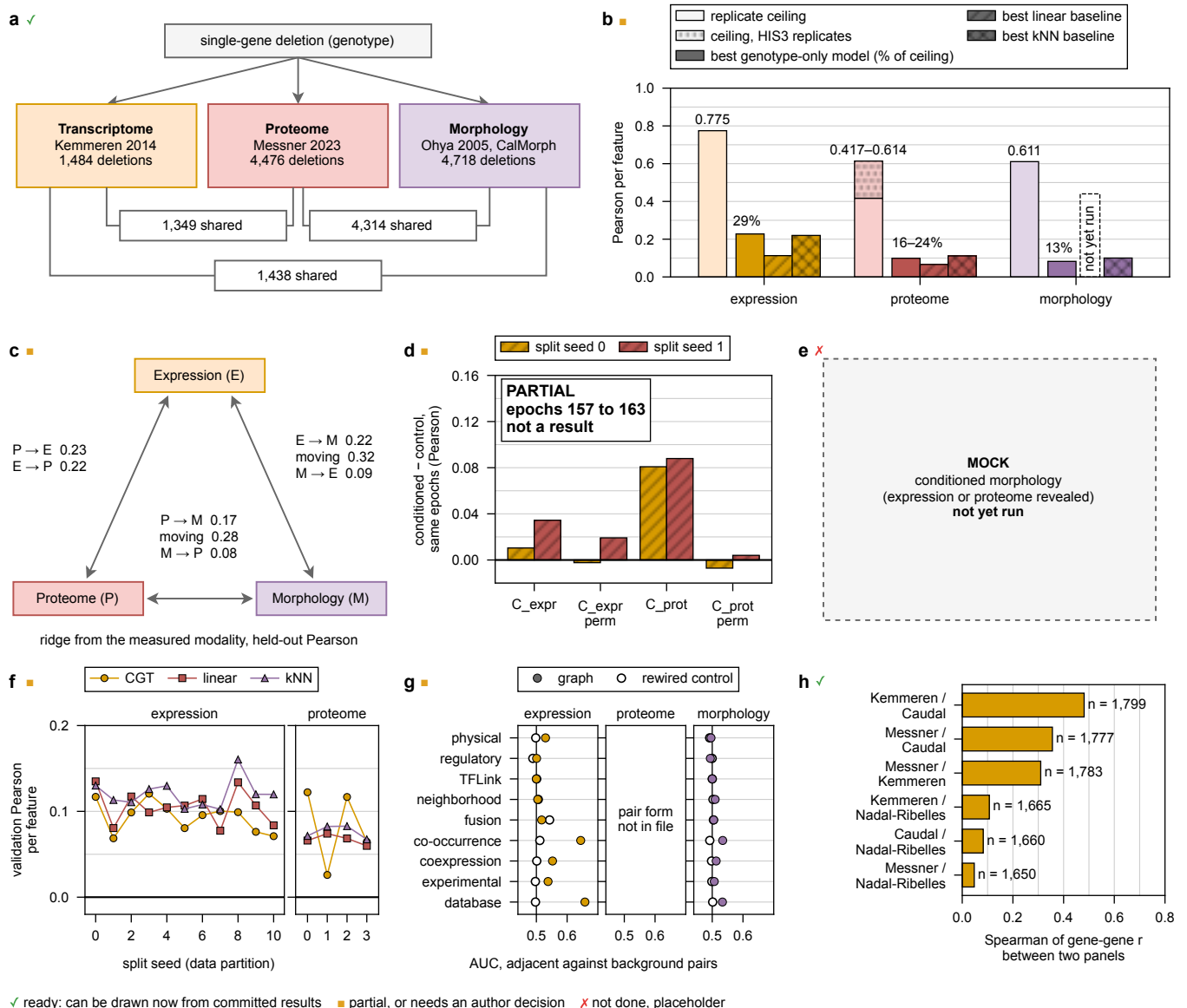
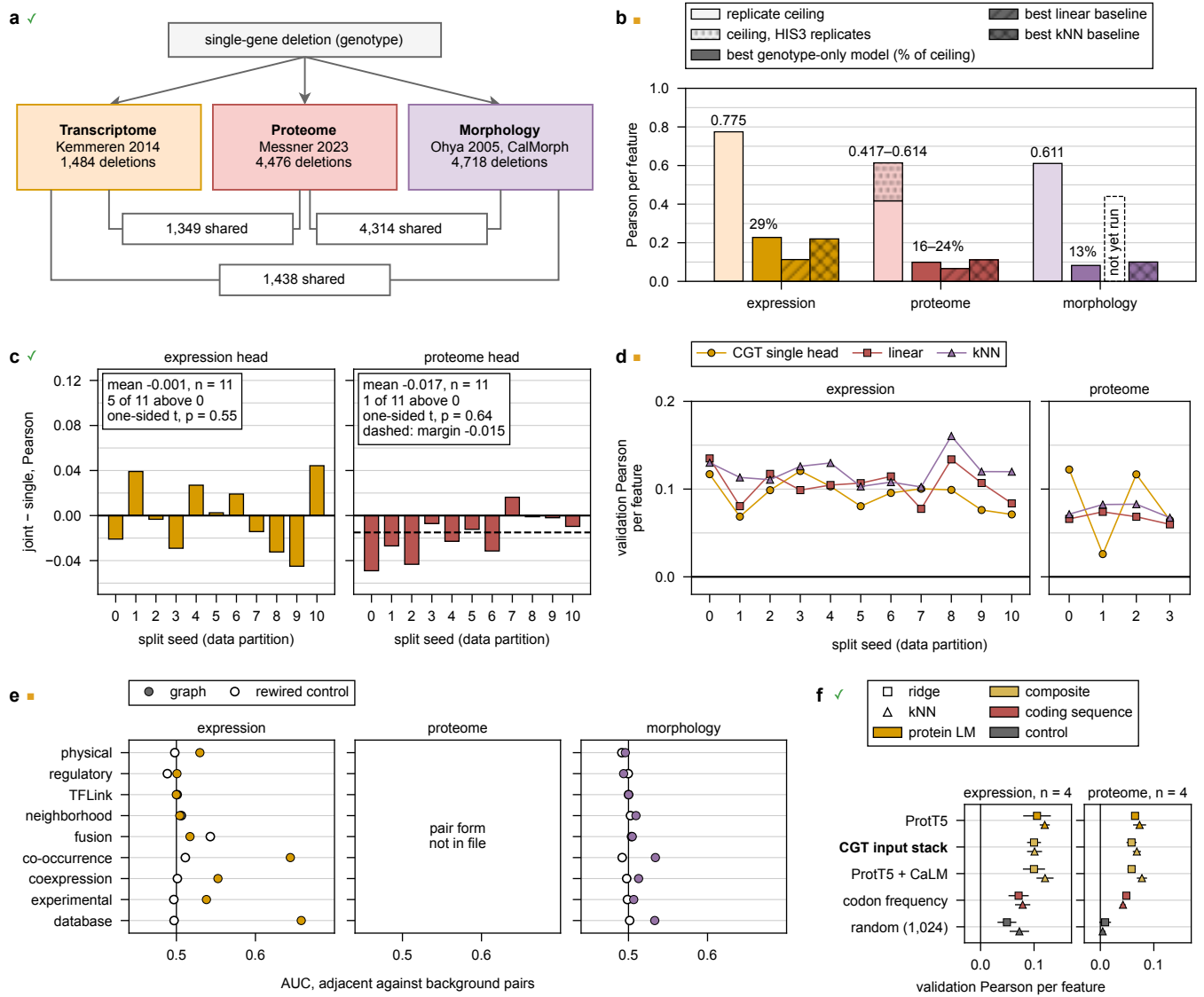


Figure 1: Ridge from one measured modality reaches a held-out Pearson of 0.17 to 0.23 on all features of another, and the conditioned model (d) has trained 160 of 1,200 epochs. **a**, The three single-deletion panels and their pairwise overlaps. Counts of distinct deletion strains; no statistic. Ready. **b**, Replicate ceiling (pale; dotted, the HIS3-replicate proteome estimator), best genotype-only model (solid; percent of ceiling), best linear baseline (diagonal hatch) and best nearest-neighbor baseline (cross hatch), as Pearson per feature across held-out strains. Model: expression and morphology are the rolling maximum of one best run over 1,353 and 183 runs, the proteome is the mean over 4 split seeds. Baselines: mean validation score over 4 split seeds of the best key (morphology, 1 split seed); no linear baseline exists for morphology. Every non-ceiling bar is a maximum over runs or keys selected on the score it receives, so it is biased upward, and the bars sit on different splits. Partial. **c**, Out-of-fold ridge between measured modalities on the deletions both panels measured (1,349, 4,314 and 1,438 strains; 5 folds): median held-out Pearson per feature; *moving* is the 116 morphology features with replicate reliability of at least 0.5. The proteome and expression side is a mean per feature from a review file. Partial. **d**, Round v20, conditioned arm minus unconditioned control: difference of the mean validation Pearson over the latest 20 epochs of each conditioned run and the same epochs of its control; 2 of 12 split seeds, 8 runs, epochs 157 to 163 of 1,200. Not a result. Partial. **e**, Conditioned morphology. Not run. Not done. **f**, Linear and nearest-neighbor baselines and the v19 single-head model per split seed (expression 11, proteome 4). Baselines: one validation Pearson per split seed, best embedding selected on that score (biased upward). Model: mean validation Pearson over the registered epoch window (unselected). A split seed is one train, validation and test partition with one initialization seed; baselines and model share the seed number and not the store, so their validation strains differ. Partial. **g**, Per graph, the AUC that two deletions adjacent on the graph give more similar phenotypes than a background pair (filled), against a degree-preserving rewired graph (open); 60,000 pairs per point, 1,482 expression and 4,718 morphology strains; no proteome strand is on file. Partial. **h**, Spearman between two panels' gene-by-gene correlation matrices, with the number of shared genes; Messner is the proteome, the others mRNA. One value per pair, no selection. Ready.

Sources for fig. 1

a	fig3_overlap_census.json (kemmeren_unique, ohya_unique); fig3_proteome_build_census.json; n_shared_strains of the two covariation files
b	expression_ceiling_replicate.json, proteome_ceiling_replicate.json, morphology_noise_ceiling.json, round_leaderboards.csv; baselines_embedding_study.json (ridge 0.113, 0.066); graph_retrieval_baseline.json (nearest neighbor 0.220, 0.112, STRING experimental graph key); knn_embedding_probe.json (0.100)
c	proteome_morphology_covariation.json (and its context block), expression_morphology_covariation.json
d	joint_checkpoint_readout.json, v20_partial.runs
f	expression_baselines_split/, baselines_split_fig3_proteome/; joint_checkpoint_readout.json, v19.runs
g	graph_prior_probe.json, pair_form, one step
h	experiments/028-knockout-expression/results/proteome_expression_covariation.json, gene_covariation

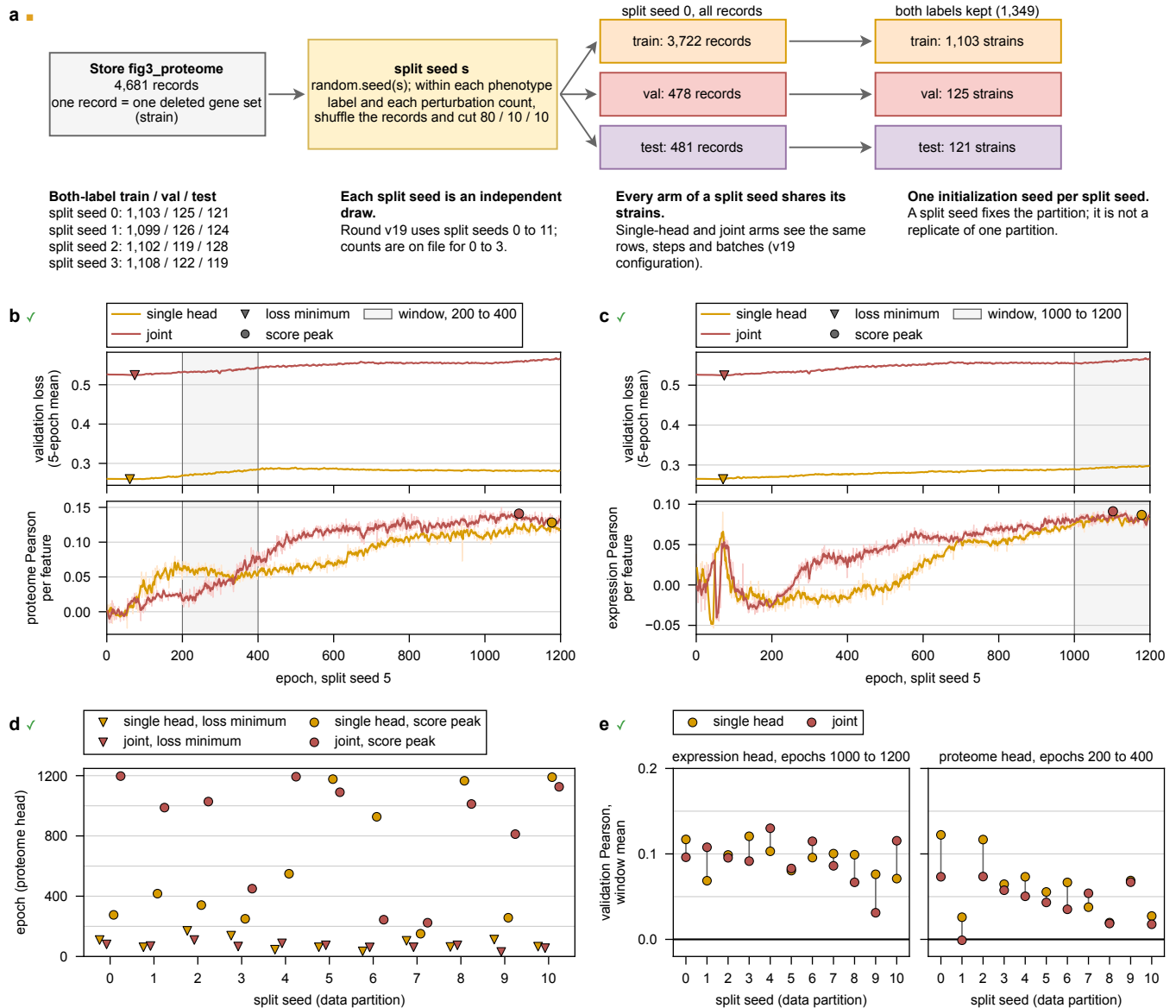


✓ ready: can be drawn now from committed results ■ partial, or needs an author decision ✗ not done, placeholder

Figure 2: From the deletion alone the model reaches 13 to 29% of the replicate ceiling, and the best nearest-neighbor baseline is within 0.01 of it on the proteome and morphology. **a**, **b**, As fig. 1a and b. Ready; partial. **c**, Round v19, joint arm minus single-head arm per split seed: difference of the mean validation Pearson per feature over the registered window (expression epochs 1,000 to 1,200, proteome 200 to 400), an unselected statistic. 11 of 12 split seeds, each a different train, validation and test partition with one initialization seed; 1,099 to 1,108 training and 119 to 126 validation strains on the four seeds on file. Expression: mean -0.001 , 5 of 11 above 0, one-sided t $p = 0.55$. Proteome: mean -0.017 , 1 of 11, $p = 0.64$ against the margin -0.015 (dashed). Neither registered test rejects. Ready; split seed 11 was unfinished. **d**, As fig. 1f at half width, same caveat. Partial. **e**, As fig. 1g at wide width. Partial. **f**, How much the gene representation matters to the two baselines: validation Pearson per feature, mean and s.d. over 4 split seeds, of the ridge baseline (squares) and the nearest-neighbor baseline (triangles), with rank and neighbor count selected on validation inside each seed. Five of 28 representations by rule, each the top of its class on the mean of the four validation means: best single language model (ProtT5), the model's input composite (bold), best other composite, codon frequency, best random control. The bold row is the study's **model_stack**, the concatenation **fudt_upstream**, **calm**, **prot_T5_all**, **fudt_downstream** that the v19 configuration inherits as **node_embeddings**. Ready.

Sources for fig. 2

- c `joint_checkpoint_readout.json`, `v19.tests` (`H1a_expression_superiority`, `H1b_proteome_noninferiority`); strain counts from `split_gene_overlap_audit.json`
 - f `baselines_embedding_study.json`; the composite is defined in `experiments/019-simb-multimodal/scripts/expression_baselines_split.py` and matched to `conf/cgt_expr_v13_split.yaml`
-



✓ ready: can be drawn now from committed results ■ partial, or needs an author decision ✗ not done, placeholder

Figure 3: In round v19 the validation loss reaches its minimum before epoch 200 on every run, and the proteome score peaks later on every run. **a**, How a split seed becomes the strains of a run, drawn from the code. The store's 4,681 records (one record is one deleted gene set) are shuffled within each phenotype label and each perturbation count and cut 80, 10, 10 under the split seed; each side then keeps the records carrying both labels, 1,349 in all, so the both-label fractions are near and not exactly 80, 10, 10. Counts for split seeds 0 to 3. No statistic. Partial: seeds 4 to 11 are not on file. **b**, Proteome head, split seed 5, one run per arm: the 5-epoch centered mean of the validation loss (top) and the validation Pearson per feature (bottom; pale raw, dark its 5-epoch mean) against epoch. Triangles, loss minimum; circles, maximum of the 5-epoch mean, an upward-biased order statistic; shaded, the registered window. The joint loss sums two heads, so the two loss levels are not comparable. Split seed 5 is chosen by rule: its joint-minus-single proteome difference (-0.012) is the median over the 11 complete split seeds. About 1,100 training and 125 validation strains. Ready. **c**, The same for the expression head of that split seed, window epochs 1,000 to 1,200. Ready. **d**, Proteome head, all 11 complete split seeds, 22 runs: epoch of the loss minimum (triangles) and of the score peak (circles). Ready. **e**, The window scores behind fig. 2c: mean validation Pearson over the registered window, single-head and joint arm, 11 split seeds, 33 runs. Unselected. Ready.

Sources for fig. 3

a	torchcell/datamodules/cell.py	(<code>_compute_and_save_index</code>);	experiments/019-simb-multimodal/scripts/train_cgt_multitask.py	(<code>require_modalities</code>); counts from <code>split_gene_overlap_audit.json</code>
b, c	joint_checkpoint_curves.csv	(41,129 rows: run, arm, split, epoch, loss, its 5-epoch mean, each head's Pearson), written by experiments/019-simb-multimodal/scripts/joint_checkpoint_readout.py; marker epochs from joint_checkpoint_readout.json		
d, e	joint_checkpoint_readout.json, v19.runs			

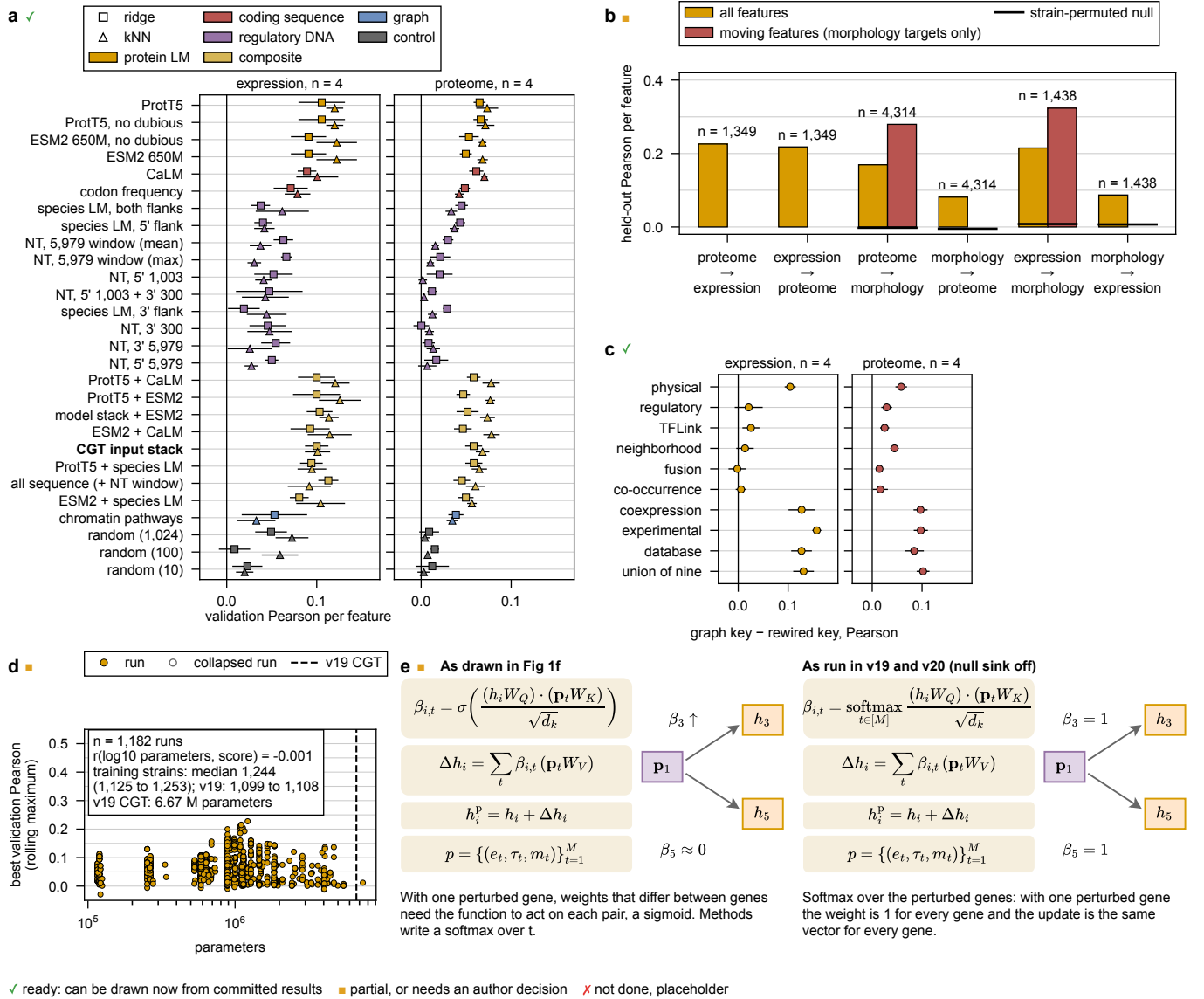


Figure 4: The model's input composite scores below ProtT5 alone in both baselines on both modalities, and the best score of a run does not track its parameter count. **a**, All 28 gene representations of the baseline study, by family: validation Pearson per feature, mean and s.d. over 4 split seeds, ridge (squares) and nearest neighbor (triangles), expression (1,244 to 1,253 training strains) and proteome (3,581); rank and neighbor count selected on validation inside each seed. Bold, the model's input composite. Ready. **b**, The six directed pairs of fig. 1c as bars with the strain-permuted null (black line, on file for four pairs) and the shared-strain count. Partial. **c**, The graph as a retrieval key: validation Pearson of a neighbor average keyed on each graph's adjacency row minus the same with the rewired graph, mean and s.d. over 4 split seeds, expression and proteome. Neighbor count selected on validation. Ready. **d**, Parameter count against the rolling maximum of the validation Pearson, one point per expression-strand run (1,182 runs with both values; open, collapsed runs). The rolling maximum is biased upward and the runs differ in epochs run. The Pearson between $\log_{10}$ parameters and score is -0.001 . Dashed, the v19 model at 6.67 million parameters for about 1,100 training strains, near 6,000 parameters per strain; its score is a window mean and is not placed on this axis. Partial. **e**, The perturbation operator in the notation of manuscript Fig. 1f; no data. Left, as drawn there. Right, as the code computes it and as the Methods write it: a softmax over the perturbed genes, which with one perturbed gene gives weight 1 to every gene. Partial, an author decision.

Sources for fig. 4

- a `baselines_embedding_study.json`
 - b as fig. 1c, plus triangle of `joint_checkpoint_readout.json` for one strain count
 - c `graph_retrieval_baseline.json`, rounds v13 and v14, `summary.paired`
 - d `round_leaderboards.csv` (`total_param_count`, `primary_roll_max`); `total_param_count` of v19.runs in `joint_checkpoint_readout.json`; `split_gene_overlap_audit.json`
 - e `torchcell/models/equivariant_cell_graph_transformer.py` (`EquivariantPerturbationTransform`); `paper/nature-biotech/sections/methods.tex`
-

3 What each panel still needs ✗

Table 1: What each panel still needs before it is real. ✓ ready: every drawn value is in a committed results file and the panel can be drawn now. ■ partial: a run is unfinished, scoring rules or splits are mixed, or an author decision is pending. ✗ not done: nothing behind it has been run.

panel		what remains
A and B, a	✓	The expression and morphology overlap reads 1,438 in <code>expression_morphology_covariation.json</code> and 1,440 in <code>fig3_overlap_census.json</code> ; reconcile.
A and B, b	■	No linear baseline for morphology is on file. The bars mix scoring rules and splits. Rescore model and baselines under one rule on one set of split seeds, and choose one proteome ceiling estimator.
A, c; SI-2, b	■	The proteome and expression side is a mean per feature from a review file, with no permuted null and no moving subset on file; recompute it with the estimator of the other two sides, on a common strain set.
A, d	■	2 of 12 split seeds at epochs 157 to 163 of 1,200. Needs the full budget, all split seeds, the registered window, and the ridge of panel c as the reference to beat.
A, e	✗	No store joins the proteome and morphology, and the observed-label channel takes gene-level values, so morphology can be a target and not yet an input.
A, f; B, d	■	Baselines and model share a seed number and not a store, and the baselines are single selected validation scores while the model is a window mean. Rerun the baselines on the both-label store. The proteome baselines cover split seeds 0 to 3 only.
A, g; B, e	■	The pair-form probe has no proteome strand; add it to <code>graph_prior_probe.py</code> .
A, h	✓	Sameith is in <code>gene_covariation_all.json</code> and not drawn.
B, c; SI-1, e	✓	Split seed 11 was unfinished at readout; redraw when it is read. The result is a null.
B, f; SI-2, a	✓	Four split seeds; the baselines ran on the full expression and proteome stores.
SI-1, a	■	Both-label counts are on file for split seeds 0 to 3 of 12. That every arm sees the same batches is stated by the v19 configuration and was not re-verified here.
SI-1, b, c	✓	One split seed by rule; the other ten are in <code>joint_checkpoint_curves.csv</code> .
SI-1, d	✓	The score peak is a rolling maximum, an upward-biased order statistic.
SI-2, c	✓	Four split seeds per strand.
SI-2, d	■	The expression document’s “145 runs, -0.129 ” is produced by no committed script and no subset of <code>round_leaderboards.csv</code> tried here reproduces it; the panel draws all 1,182 scored expression runs, $r = -0.001$. Scores are rolling maxima over unequal budgets. A v19 point needs a v19 score under the same rule.
SI-2, e	■	Author decision: Fig. 1f draws gene-dependent weights with one perturbed gene, and the Methods and the code compute a softmax, which gives weight 1. Either the figure or the operator changes. The null sink, which restores a sigmoid weight, has one trained comparison stopped at 186 to 189 epochs.

4 What the audit of 2026-10-04 established ✗

Ten independent read-only reviews of the expression, proteome and morphology work were run on 2026-10-04, one scope each, and are filed under `review/2026-10-04/`. Where two reviews reached the same point from different files it is stated once. Every number is measured unless it is marked as a hypothesis.

No trained model on file beats the best baseline on matched strains. On the test strains of the model’s own partitions, read at the best-validation checkpoint, the graph transformer scores 0.110 and 0.153 on expression (split seeds 0 and 2) against 0.165 and 0.174 for a nearest-neighbor average keyed on the interaction graphs, and 0.067 and 0.082 on the proteome against 0.117 and 0.111. The model minus the best validation-selected key is negative in six of six comparisons, by 0.02 to 0.06. This rests on two split seeds per modality, because test predictions were saved for six of forty checkpoints; validation orders the same way on all four split seeds. Morphology has no matched comparison at all. A second review reads a different set of checkpoints (v17 reference arm, three seeds per split) at a model edge of about $+0.017$ over the matched sequence baselines, most of it from split seed 0, so the sign against sequence baselines is unsettled and the sign against the graph key is not. SRC: `review/2026-10-04/04_baselines_and_ceilings.md,07_data_splits_metrics.md`

Headline numbers that do not hold. The incumbent’s 0.196 ± 0.022 is the mean of eight different arms of one round, all at seed 0 on split seed 0, scored by a rolling maximum; same-seed reruns agree within 0.001, and no configuration has been replicated at a long budget. Split seed 0 is a favorable draw: its validation score sits 1.72 standard deviations above the mean of twelve baseline draws, and every long run used it. The claim that the model clears the bilinear baseline by 0.093 compared a rolling maximum against a baseline scored on a different permutation of the strains. The rank-64 pair term’s advantage per unit of compute compared one run in each of two projects and fails at matched budgets. The statistic “145 runs, correlation of log parameters with score -0.129 ” has no generating script and is not reproduced by `round_leaderboards.csv`, which gives -0.001 over 1,182 runs confounded with budget. The manuscript’s Results still carry placeholder correlations of 0.543 and 0.619 for expression and morphology. SRC: `review/2026-10-04/01_expression_pre_simb.md,02_expression_post_simb.md,06_training_dynamics.md,09_prior_reviews_and_literature.md`

What has a measured effect. Budget, within a single run: 0.147 at epoch 500, 0.202 at 2,000, 0.230 at 4,000. The

partition: between-partition standard deviation 0.020 to 0.030 against 0.010 to 0.020 within, larger than any arm effect. The masked objective: removing it costs 0.012 over twelve pairs with a confidence interval that excludes zero, negative on all four partitions; rounds v19 and v20 run without it. Embedding content against width-matched random vectors: +0.068. Interaction graphs as a neighbor key against their rewired controls: +0.150 on expression and +0.115 on the proteome, four of four split seeds. Among sequence representations ProtT5 alone is at or above every composite, and the four-part stack the model consumes adds -0.03 to $+0.003$ over it. No architecture change has met the campaign's own bar of $+0.02$ on three of four partitions with the test sign agreeing. SRC: [review/2026-10-04/01_expression_pre_simb.md](#), [02_expression_post_simb.md](#), [04_baselines_and_ceilings.md](#)

What the code does at one deletion. The v19 model has 6,681,548 parameters, of which 5.69 million, 85 percent, are the first linear layer that projects the 3,328 dimensional concatenated embedding; the encoder is 0.59 million and the perturbation operator 0.10 million. The encoder runs once on the wild-type graph, so its output is the same for every strain. The operator is a softmax over the perturbed genes; with one deleted gene its weight is identically 1 and every gene receives the same vector, so its query and key parameters (16,380) receive zero gradient. The Methods describe this softmax; Fig. 1f draws a weight that differs between genes, which this operator cannot produce at one deletion. The operator's attention dropout acts on that single weight and drops whole heads of the deletion signal at random. Nothing marks which gene was deleted: the model predicts $+0.036$ for the deleted gene's own transcript against a measured -2.42 , and a predictor that writes only that one entry scores 0.0175 per-feature Pearson, the size of the edge in dispute. The null sink, the sigmoid form that makes the weight depend on the querying gene, was trained once, for 186 to 189 epochs, with its gate bias near -4 . SRC: [review/2026-10-04/05_architecture_and_operator.md](#), [07_data_splits_metrics.md](#), [01_expression_pre_simb.md](#)

What the training curves show. The runs are neither converged nor memorizing. Train-side Pearson is still rising at epoch 1,200 and flattens below 0.78 after 10,000 epochs. The validation-loss minimum near epoch 70 to 90 sits at the end of a plateau where predictions are still near-constant, so it marks the launch of the head, not the onset of overfitting, and the loss-minimum checkpoint is close to a mean predictor. Predictions are over-dispersed for their correlation (spread ratio about twice the Pearson), which is why the normalized error exceeds 1; a term that pushes the prediction spread toward the target's would raise that error further. Wall time does not depend on model size: an epoch costs the same at 0.68 and 1.45 million parameters, a fifth of it is the eval-mode train pass, and the encoder is recomputed every step although its output never changes. With the encoder output cached a CPU step falls from 42 s to 2.6 s, and to 0.23 s on 512 reporter genes. SRC: [review/2026-10-04/06_training_dynamics.md](#), [05_architecture_and_operator.md](#)

Defects found. Three bear on numbers already reported. The prediction-dump routine calls the model without the observed-label input, so every dumped validation and test prediction from the masked-objective lineage comes from a different function than the logged metric. The v16 expression baselines were taken from the wrong store's partitions. The per-feature Pearson drops near-constant columns per model, so two arms can be scored on different feature sets. One bore on running jobs: the conditioned arms were launched without persistent data-loader workers, the setting whose absence killed four v15 runs; the pending v20 tasks were resubmitted with it (job 2423297) and tasks 0 and 1 of job 2423190 run as launched. SRC: [review/2026-10-04/05_architecture_and_operator.md](#), [07_data_splits_metrics.md](#), [03_proteome_and_joint.md](#)

What new data can and cannot do inside the window. The local SPELL archive holds 200 studies with genetic perturbations covering 1,175 genes, of which 478 are not among the 1,484 Kemmeren deletions: 216 are promoter knockdowns of essential genes, 146 come from Hughes 2000, and 14 from Hu 2007. On the ridge learning curve 245 to 460 more strains project to about $+0.004$ to $+0.008$ (hypothesis, one split), below the between-partition spread. The natural isolates need a reference record, per-allele sequence embeddings that no builder supports, and a loader fix, five to eight days before a first number. The two amino-acid screens do not agree with each other because the Cooper screen does not reproduce its own duplicate strains. No relationship between fitness and the expression or proteome profile has been measured. SRC: [review/2026-10-04/08_cross_modal_and_new_data.md](#)

The conditioned round. The conditioning step is wired as intended: validation and test are conditioned, standardization is fit on train, the revealed head leaves the loss and the metrics. Its comparison against the v19 single-head arms is not like for like (head slot, loader setting, and an encoder input of values against zeros), so the primary contrast is each conditioned arm against its permuted control, and the bar is the ridge from genotype plus the measured modality refit per partition (0.236 on expression and 0.214 on the proteome on one split), not the unconditioned arm. SRC: [review/2026-10-04/03_proteome_and_joint.md](#)

5 The plan to the deadline ✗

Fewer than twenty days remain. The plan is ordered so that the first week produces a figure that exists whatever the later experiments show, and so that each later line of work has a date on which it is dropped.

Days 0 to 4, almost no GPU: the benchmark that cannot fail to exist. Four pieces. First, the linear and nearest-neighbor baselines for every gene representation on exactly the strains rounds v19 and v20 use, twelve split seeds, both labels (IGB jobs 2423298 to 2423302, running). Second, one scoring harness applied to everything already trained: predictions dumped through the same function the metric uses, the deleted gene’s own entry blanked, a feature set fixed in advance by replicate reliability, window means in place of rolling maxima, test strains, a strain bootstrap. Third, a harness that tells the candidate limits apart: a ridge and a bilinear model on 137 to 1,100 training strains over five splits, and the same on planted labels at the measured noise level. Fourth, two gates that earlier reviews asked for and no one ran: perturbation-derived embeddings of the deleted gene (its genetic-interaction, chemogenomic, morphology or proteome profile) through the existing baselines, and measured fitness as an input to the genotype ridge with a shuffled control. **Gate, day 4:** the matched table is drawn with ceilings. If the model is below the graph key on at least three of four split seeds per modality, Figure 3 reports a benchmark, and the remaining work is to raise the model to the graph key.

Days 2 to 8: many small models. The encoder output is cached, so only the operator and the heads train, on the both-label store already on IGB. Arms, each against a paired reference in the same pack: the sigmoid-gated operator with an explicit term in the reporter and the perturbation and an indicator of the deleted gene; the deleted gene’s graph neighborhood as input; a projection cut from 5.69 million parameters to a frozen low-dimensional one; ProtT5 alone against the stack; the masked objective on; the perturbed CLS as the morphology readout once the 30-line port from the interaction branch is in. Screen at 300 epochs on three split seeds, scored by the mean over epochs 200 to 300. Hypothesis (untested): the cached model runs at least ten times faster per epoch on a GPU. **Gate, day 8:** the operator line is dropped unless an arm beats its paired reference on at least nine of twelve split seeds at confirmation and reaches the graph key.

Days 8 to 15: confirmation. At most two winners and the reference at full size on twelve split seeds with a sealed test set, about 2.5 days on three A40s.

The conditioned round, in parallel. It runs to completion on the two cabbi cards. **Gate, about day 10:** with at least six split seeds, a conditioned arm must exceed its permuted control by 0.02 and reach the per-partition ridge; otherwise it is reported as a model that uses a second modality less well than a linear map.

What is not in the window. New deletion loaders from SPELL, unless the day-one learning curve projects a gain above 0.01. Natural isolates, unless the dataset-build machine is back by day 5. The distribution-matching loss term, the dataset token, and any pooled single-cell screen.

Days 16 to 19. Freeze, read the sealed test set once, draw the figure, write.

Decisions that are the author’s. Whether Figure 3 may be a benchmark in which a graph neighbor average ties or beats the transformer. Whether Fig. 1f or the Methods changes, since they name different operators. Whether the manuscript’s placeholder correlations and its statement that the model predicts expression and morphology jointly are removed now. Whether the pending v20 tasks keep both directions.